

Exploring Deep Learning Tools: A No-Code Introduction to TensorFlow and Keras Reflective Journal

This lab gave me a clearer understanding of how a pre-trained deep learning model can be used without having to build or train the model from scratch. Before completing the lab, I understood the general idea that convolutional neural networks are used for image classification, but I had less familiarity with what the model architecture actually looks like and how an image must be prepared before the model can make a prediction.

One of the most useful parts of the lab was reviewing the VGG16 model architecture. The model summary showed how an image moves through multiple convolutional and pooling layers before reaching the fully connected layers used for classification. I observed that VGG16 expects an input image size of 224×224 pixels with three RGB color channels. I also saw how the spatial dimensions of the image become smaller as it moves through pooling layers, while the number of feature maps increases. This helped me understand how the network gradually transforms raw image data into more complex visual features.

The preprocessing step was also important because it demonstrated that an image cannot simply be passed into the model in its original form. The image had to be resized, converted into a numerical array, given a batch dimension, and processed using the format expected by VGG16. This made the relationship between data preparation and model performance more concrete for me.

I especially enjoyed testing the model with my own images. When I used a photo of my cat Gigi, the top predictions included **tiger_cat** and **tabby**, which were reasonable classifications. However, when I tested a photo of my other cat, Henry, the model predicted **printer**, **refrigerator**, and **dishwasher** as the top results, which I found quite humorous. Even though both photographs clearly showed cats to me, the model interpreted the images very differently.

That comparison was probably the most valuable part of the lab for me. It demonstrated that a deep learning model does not understand an image in the same way a person does. Instead, it relies on visual patterns and features learned during training. Differences in pose, background, composition, and preprocessing can influence which features the model emphasizes and therefore change the prediction results.

Overall, this lab helped me better understand the workflow of a pre-trained image classification model, including model architecture, preprocessing, prediction, and interpretation of results. It also reinforced the importance of evaluating model outputs critically rather than assuming that a prediction is correct simply because it was generated by a deep learning model. I now feel more comfortable with the basic process of using a pre-trained CNN and interpreting its predictions.